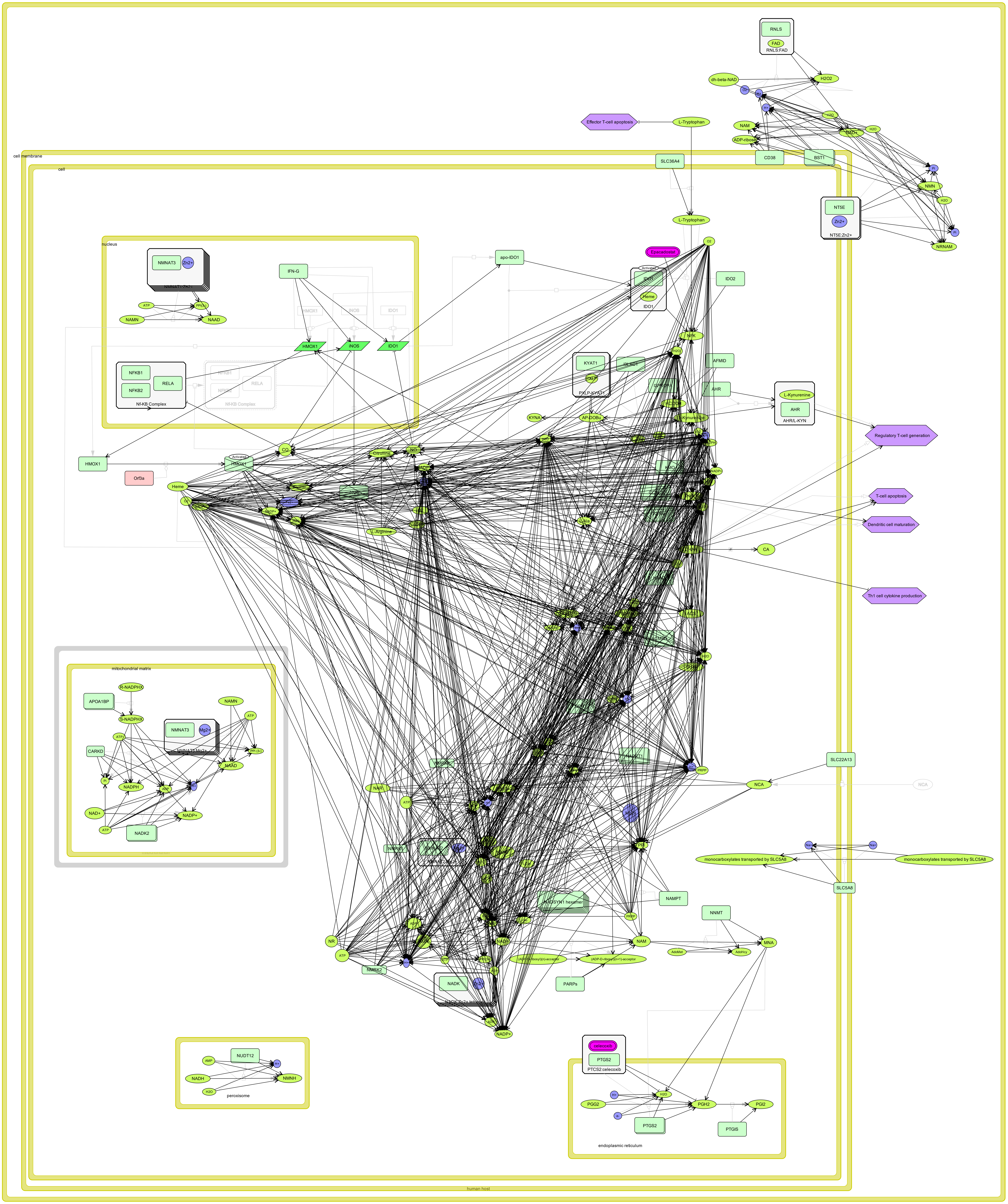
RNLS
FAD
RNLS:FAD
dh-beta-NAD
H2O2
O2
H+
H+
H2O
H2O
NAM
NAD+
ADP-ribose
CD38
BST1
Pi
NMN
H2O
NT5E
Zn2+
NT5E:Zn2+
Pi
NRNAM
Effector T-cell apoptosis
L-Tryptophan
cell membrane
cell
SLC36A4
L-Tryptophan
nucleus
NMNAT3
Zn2+
NMNAT1:Zn2+
ATP
PPi(3-)
NAMN
NAAD
IFN-G
HMOX1
iNOS
IDO1
HMOX1
iNOS
IDO1
apo-IDO1
Epacadostat
O2
Activated
IDO1
Heme
IDO1
IDO2
NFKB1
NFKB2
RELA
Nf-KB Complex
NFKB1
NFKB2
RELA
Nf-KB Complex
NFK
H2O
KYAT1
PXLP
PXLP-KYAT1
I3LACT
AFMID
I3PROPA
AHR
L-Kynurenine
AHR
AHR/L-KYN
HCOOH
KYNA
AP-DOBu
L-Kynurenine
Regulatory T-cell generation
CO
Citrulline
NO
HMOX1
Activated
HMOX1
Orf3a
Heme
O2
NADP+
Fe2+
NADP+
H2O
iNOS
L-Arginine
NADPH
NADP+
KMO
L-Ala
T-cell apoptosis
Dendritic cell maturation
CA
Th1 cell cytokine production
ACS
NADH
H+
ACMSD
H2O
QUIN
mitochondrial matrix
R-NADPHX
APOA1BP
S-NADPHX
NAMN
ATP
ATP
NMNAT3
Mg2+
NMNAT3:Mg2+
CARKD
PPi (3-)
NAAD
Pi
NADPH
ADP
H+
NAD+
NADP+
ATP
NADK2
mitochondrion
NMRK2
NAR
ATP
H+
NAPRT1
H+
PRPP
SLC22A13
NCA
NCA
NMRK1
NMNAT2
Mg2+
NMNAT2:Mg2+
Mg2+
PPi
Na+
Na+
monocarboxylates transported by SLC5A8
monocarboxylates transported by SLC5A8
SLC5A8
Activated
NADSYN1 hexamer
NAMPT
PRPP
L-Glu
NNMT
NR
ADP
NAD+
NAM
MNA
ATP
H+
NMRK2
ATP
PPi(3-)
ATP
AdoMet
AdoHcy
(ADP-D-ribosyl)(n)-acceptor
(ADP-D-ribosyl)(n+1)-acceptor
PARPs
NADK
Zn2+
NADK:Zn2+ tetramer
ADP
NADP+
AMP
NUDT12
H+
NADH
NMNH
H2O
peroxisome
celecoxib
PTGS2
PTCS2:celecoxib
H+
H2O
PGG2
PGH2
PGI2
e-
PTGS2
PTGIS
endoplasmic reticulum
human host

H2O
CD38
NT5E
Zn2+
NT5E:Zn2+
Orf3a
BST1
NUDT12
AMP
H2O
NADH
PARPs
(ADP-D-ribosyl)(n)-acceptor
SLC22A13
SLC5A8
monocarboxylates transported by SLC5A8
Na+
RNLS
FAD
RNLS:FAD
O2
dh-beta-NAD
H+
H+
NMNH
AdoMet
NNMT
NAM
ATP
QPRT
NMRK1
NMRK2
NR
NAR
NCA
Na+
NAPRT1
monocarboxylates transported by SLC5A8
Mg2+
NAMPT
PRPP
AdoHcy
MNA
H+
PGG2
PTGS2
celecoxib
PTGS2
PTCS2:celecoxib
e-
NMNAT2
Mg2+
NMNAT2:Mg2+
NMN
NAMN
H2O2
NAD+
NMN
ADP-ribose
NAM
PTGIS
PGH2
H2O
L-Gln
Activated
NADSYN1 hexamer
NAAD
HMOX1
PPi(3-)
NRNAM
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PGI2
NAD+
iNOS
L-Arginine
Activated
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NADPH
O2
Heme
apo-IDO1
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Zn2+
NADK:Zn2+ tetramer
SLC36A4
L-Tryptophan
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NMNAT3
Zn2+
NMNAT1:Zn2+
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RELA
NFKB2
NF-KB Complex
IFN-G
Biliverdin
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Fe2+
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HCOOH
AHR
L-Glu
AMP
NADP+
KYNU
3HKYN
CCBL2
I3LACT
PYR
KYAT1
PXLP
PXLP-KYAT1
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Mg2+
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NADPH
Pi
ADP
H+
NADP+
NAAD
PPi (3-)